FedCast

A forecaster for the next Federal Reserve rate decision, built to be judged by its process rather than its luck

Tan Kah Ming · Fullerton · October 2026

About me



Tan Kah Ming

Senior AI, Data and Technology Consultant, seconded to Sovereign Wealth Fund.

Before that

- Data scientist, RSAF Data Office: contributed to RSAF AI Strategy, set up its first digital unit
- Air engineering and weapons logistics officer, where the data-driven habit started (Weibull analysis, predictive maintenance)

Training

M.Eng Aeronautical Engineering, Imperial College London · Graduate Certificate in Predictive Analytics, NUS · CFA Level 3 candidate (February 2027)

Things I have built that models/Applications in

1

Investment analytics, valuation and execution

Investment analytics: AI-assisted performance attribution and idea tracking for an investment team.
Valuation: Machine-learning valuation for illiquid assets.
Execution: Trade workflow automation for investment teams.

2

Data Science

- 4th of 160+ teams (MIT, DoD, USAF) in the 2023 USAF AI Accelerator, predicting pilot flying difficulty from physiology;
- An ammunition recommender (quantile LightGBM) that cut wastage with no shortfall -> Defence Manpower Committee Innovation Award 2022.

3

COVID-19, 2021

Built, in weeks, the system that engaged every home-recovery patient in Singapore, 10,000 people a day, on form.sg, Azure and Zendesk. The lesson that carries over: a system people rely on must say what it knows and what it does not.

The question, the answer, and what happened

28 Oct

2026: the decision being forecast

67.6%

FedCast, no change: ledger entry #18, 27 hours before

Hold

what the Fed did: 3.75 to 4.00%, unchanged

10 to 2

two dissents, both for the +25 bps the specialists leaned toward

How the number was made

- One frozen snapshot of 23 evidence items, pulled 28 October: futures, eight FRED series, both statements, minutes, transcript, projections, speeches, 32 years of decisions, the NY Fed natural rate, prediction-market odds
- Four models pooled by code at fixed weights (market 0.6, Taylor rules 0.2, odds 0.1, base rates 0.1), then a 1.5% floor on every outcome: no change 70.9%, hike 26.2%
- Five specialist analysts, every claim quoted and verified by code, lean the number +4.0 bp toward a hike: no change 67.6%, +25 bps 22.4%, +50 bps 7.7%
- Hash-chained ledger entry #18; recomputes exactly from its snapshot and the analyst runs it names

What the outcome says

- The most likely outcome was the outcome. The market alone was more confident (90/10); at five weeks we do not claim to beat it
- What the system added was the reasons: the hawkish lean came from the communications, minutes and inflation readers; the labour reader was the lone dove. Two dissents for a hike is that lean, in the vote
- Every reason is a quote code found in the frozen evidence, so the call can be taken apart claim by claim, which the rest of this talk does
- One decision proves little. It is scored formally (log score, Brier, against the market and two naive baselines) the moment the NY Fed's post-meeting rate row lands, and every weight decision waits on that record

Live demo 1: the forecast

1 min

Open

- The app at localhost:8765, Forecast view
- Top bar: when the evidence was pulled, what is new at the sources

Point at

- The two tracks side by side: machine only, and with views
- The meta line: ledger entry, snapshot hash, the specialists' tilt
- The five bars

Say

- This number was made at <time> from evidence frozen at <time>. Everything I show next is how it got there.

LIVE DEMO

Why build this

1 We judge the process, not the luck

- Eight decisions a year: a good year proves nothing
- The yardstick is a ten-point scorecard, fixed before any code: same evidence, same answer? every claim quoted? any number from memory? a shock moves the number the right way?
- Outcomes are logged after each decision and never used to tune

2 Every step is auditable

- An analyst that says 'inflation remains elevated' must quote the statement line it read, and code checks the quote is there, character for character
- Every number it uses must exist in the frozen evidence, or it is flagged as from memory
- Every model lists the items it read; every ledger entry recomputes from its snapshot. If it is not quoted, it does not count

3 A harness that compounds

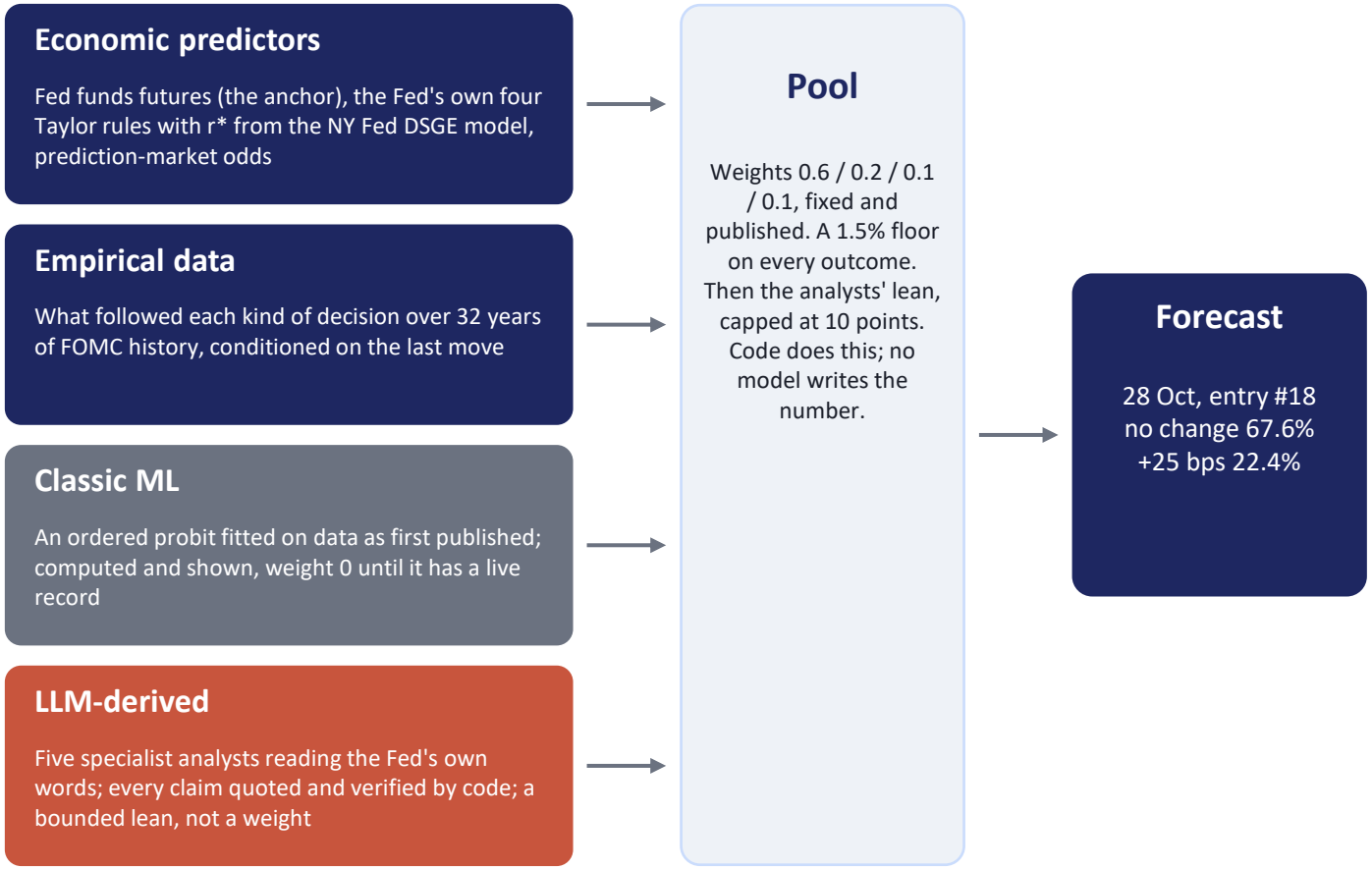
- Everything is quotable and the analysts can read only the evidence, so a wrong call is a traceable wrong call: we see which quote led where
- The harness is repeatable, so it goes wrong the same way every time; a fix can be tested, and the drift table shows it stayed fixed
- That is the virtuous cycle: audit, fix, re-measure. The ladder, the scorecard and the drift table later in the deck are that cycle

What others do, and what each lacks

Predictor	What it is	Strength	What it lacks
Fed funds futures (CME FedWatch)	Price of the contract that settles on the month's average rate	About 88% right at 30 days over 1994 to 2024 (Bonini, Huang, Simaan). The anchor of our pool	No reasons. One number with no evidence trail; fails silently when the Fed surprises
Prediction markets (Polymarket, Kalshi)	Crowd odds on each outcome	Independent crowd; fast to move	Thin and fee-distorted; a $22 entry can sit next to a $67m one
Economist surveys	Median of polled forecasters	Human judgement, broad	Slow, unauditable, consensus-seeking
Policy rules (Taylor family)	Rate a formula says is appropriate	Transparent, the Fed's own benchmarks	Depends on unobservable r* and u*; says where rates should be, not what the committee will do
LLM 'FOMC agent' papers (2025)	Models that read Fed text and vote	Reported 90%+ accuracy on 2023 to 2024	16 meetings; no frozen evidence, no quote verification, no replay, no control
FedCast	All of the above pooled by code, plus verified analysts	Every number traceable; repeatable; a scorecard that catches drift	No live record yet; no accuracy claim at five weeks

Sources: Bonini, Huang and Simaan, 'Watching the FedWatch' (232 meetings, May 1994 to March 2024); FedSight AI preprint, December 2025; Takano, Hirano and Nakagawa, arXiv 2511.02469.

Four families, one pool, one number



The chain, in order

Evidence	23 items frozen and fingerprinted, dated in US time, data as first published. Nothing outside it may be read.
Signals	Computed by code and handed to the analysts as numbers: inflation gap, labour slack, financial conditions, statement tone, base rates, projections.
Pool	Four families at fixed weights, the floor, then the specialists' lean.
Analysts	Only analysts 4 to 8 enter the number. Analysts 1 to 3 are checks and balances on the ladder.
You	Views tilt by at most 10 points and expire; watch-outs must be answered by id; documents and odds become evidence.
Ledger	Append-only, hash-chained; every entry recomputes from its snapshot and the runs it names.

The arithmetic: from five models to one number

Model	What it reads	–25	no change	+25	+50	weight
Market-implied	November fed-funds future against the overnight rate: +2.5 bp expected, split between the two nearest outcomes	0	90	10	0	0.6
Taylor family	12-month core PCE 3.0%, unemployment 4.2%, r* from the NY Fed DSGE model (0.65 / 1.86 / 3.06%), four rules × nine r*/u* pairs: +22.5 bp mean	4.5	26.9	44.6	23.8	0.2
Prediction market	Polymarket, 26 Oct, $67m traded: 94.5% no change	0.2	94.9	4.5	0.4	0.1
Historical base rates	What followed a hike at the 49 scheduled meetings since 1994 that came after one	0	34.7	52.2	13.1	0.1
Ordered probit v2	Nine features on first-published data; failed v1, caveated v2	0	24.2	63.8	12.0	0
Pooled	Weighted average, then the 1.5% floor	**1.5**	**70.9**	**20.2**	**6.0**	
Specialists' lean	+4.0 bp hawkish, mean of analysts 4 to 8: 4.0 points moved toward the hike outcomes					
Machine forecast	Ledger entry #18, 28 October	**1.2**	**67.6**	**22.4**	**7.7**	

The weights are judgement, logged, not backtested. The backtest (expanding window, 129 decisions since 2010) decides who is in the pool: the base rates beat the fitted probit and joined; the probit failed its gate and sits at weight 0. Live scores from 28 October onward decide the weights, by amendment, on the record. Fitting them on the same history the models came from would prove nothing.

Live demo 2: the evidence graph and the pool

3 min

Open

- Evidence graph
- Click the Market-implied box
- Click the Pool box
- Methodology, scroll to the worked example

Point at

- Market-implied: the futures price, 100 minus price, the expected change, the 90/10 split
- Pool: the four weights, the pooled line, the specialists' tilt line, machine_only
- Any evidence item: the frozen text and the link to the original

Say

- Blue is in the number, grey is computed but not in it. Click a model and you see only what it read.
- The language model's lean is the one line between pooled and machine_only. It can tilt, it cannot write.

LIVE DEMO

Eight analysts, one model, one ladder

1	**Analyst 1: naive** The question and the date. Nothing else.	What the model does on its own. The control every later step is measured against.
2	**Analyst 2: evidence shown, nothing verified** Every document and every pre-computed number, 128,000 characters. Free-form answer.	Adds the evidence. Shows what evidence alone fixes, and what it does not.
3	**Analyst 3: generalist with verifier** Same evidence, but a fixed answer shape: a verbatim quote per claim, a watch-out report, a tilt capped at ±10 bp. Code checks every quote and number.	Adds structure and verification. This is where the output becomes stable and checkable.
4-8	**Analysts 4 to 8: specialists** One slice each: communications, minutes, inflation, labour, financial conditions. Same schema, same verifier.	Adds specialisation. Each judgement is attributable to its evidence; a dissent survives instead of being averaged away.

Each step adds one thing; the value of that thing is the difference in score between neighbouring steps. Only analysts 4 to 8 enter the number. Analysts 1 to 3 never touch it: they exist as checks and balances, to show what the harness adds and to catch drift.

What the harness enforces around every analyst

1	**Tools off, slice only**	The model sees its slice of the frozen snapshot and nothing else: no browsing, no memory of the Fed, no other analyst's output.
2	**Pre-reads**	Numbers are computed by code and handed over: inflation gap, labour slack, financial conditions, tone, base rates. The model never does arithmetic.
3	**Fixed answer shape**	Claims, each with a verbatim quote and the item it came from; a report on every watch-out by id; a tilt bounded to ±10 basis points. Enforced by the runtime.
4	**Code verifies**	Every quote is string-matched to its item. Every number is matched to the snapshot. Every watch-out must be answered. Failures are counted, never hidden.
5	**Recorded and repeated**	Five identical runs per analyst per snapshot, each stored with its prompt hash. Repeatability is measured, and a recorded run replays for free.
6	**One repair round**	If a quote is not found or a number is not in the evidence, the answer goes back once with the verifier's report: fix the quote verbatim, drop the number, or drop the claim. Both answers are kept; the repaired one counts. No second round.

Live demo 3: what each analyst is asked and given

2 min

Open

- Analyst prompts view
- Analyst 1: open the full prompt
- Analyst 6: open the full prompt

Point at

- Analyst 1: 195 characters, no evidence
- Analyst 6: the system prompt, the enforced answer schema, the inflation pre-read, the item ids
- The prompt hash: five repeats share it

Say

- If it is not on this page, the analyst does not get it.
- Same model. The only difference between analyst 1 and analyst 6 is this page.

LIVE DEMO

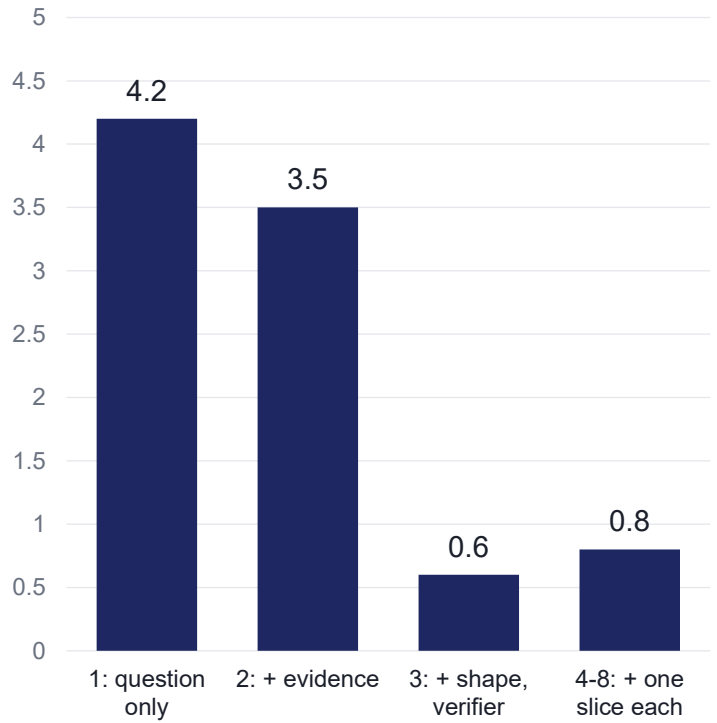
The ladder, four snapshots, five runs each

Dimension (pass mark)	Analyst 1: the question only	Analyst 2: + the evidence	Analyst 3: + answer shape + verifier	Analysts 4-8: + one slice each
S1 Repeatability (≤ 2 pp) max move of any probability across 5 identical runs	**3.4 / 4.5 / 4.9 / 4.1 pp, fail**	**3.3 / 2.3 / 4.8 / 3.4 pp, fail**	**0.6 / 0.5 / 0.5 / 0.8 bp of tilt, pass**	**0.5 / 0.8 / 1.2 / 0.8 bp of tilt, pass**
S2 Groundedness (≥ 95%) share of claims whose quote code found in the evidence	**0%, fail**	not measurable: no quotes	**100 / 100 / 100 / 99%, pass**	**100 / 97 / 100 / 100%, pass**
S3 Leakage (exactly 0) numbers in the answer that are not in the snapshot	**36 / 36 / 39 / 38, fail**	**0, pass**	**0, pass**	**0, pass**
What it concluded	Cut 25 bps at 57 to 67%: its memory of the Fed is stale	+25 bps at 50 to 57%: reads the evidence, overshoots the market	+7.0 / +6.6 / +6.4 / +5.8 bp hawkish	+4.4 / +4.1 / +4.4 / +4.0 bp hawkish; labour the lone non-hawk

Snapshots of 15, 24, 26 and 28 October 2026. Analysts 3 to 8 report S1 in basis points of tilt because they output a lean, not a distribution.

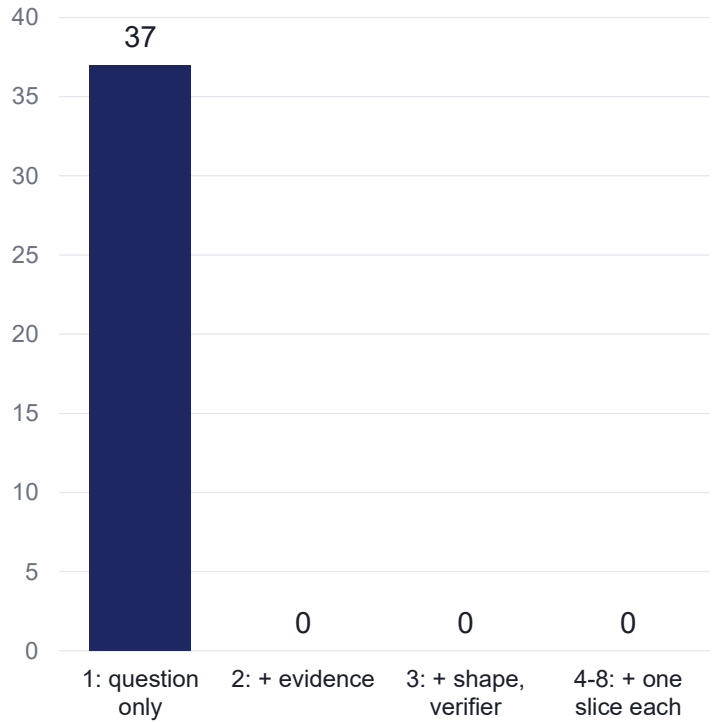
How good is the harness: what each step bought

S1: wobble between identical runs (pp)



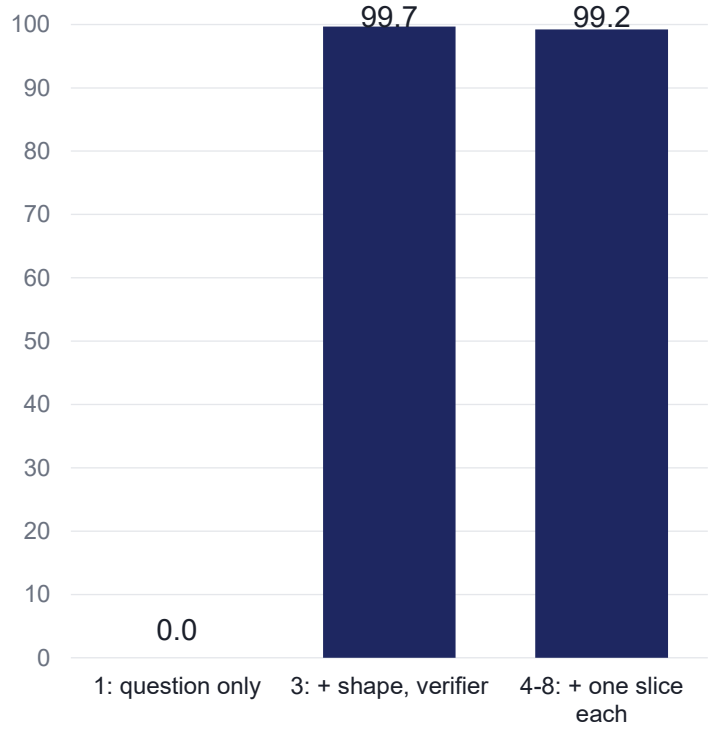
Largest standard deviation across five runs, mean of four snapshots. Pass mark 2 pp. Analysts 3 to 8: standard deviation of the lean, 1 bp = 1 pp.

S3: numbers from memory (count)



Numbers in the answer that exist nowhere in the frozen evidence, summed over five runs. Pass mark: exactly 0.

S2: claims with a verbatim quote (%)



Share of claims whose quote code found character for character in the item named. Pass mark 95%. Analyst 2 is not shown: it quotes nothing, so it cannot be measured.

The step from analyst 2 to analyst 3 is the harness: a fixed answer shape with a quote per claim, checked by code. It is the step that makes the model repeatable and checkable. Specialists keep both and add attribution.

What the specialists said on 26 October

Analyst	Reads	Lean (5 runs)	Verified
4 Communications	Both statements, minutes, transcript, speeches, tone pre-read	**+6.0 bp hawkish**	100%, 0 leaked
5 Minutes digest	The minutes only	**+6.6 bp hawkish**	100%, 0 leaked
6 Inflation	Inflation pre-read, projections, both statements	**+6.4 bp hawkish**	100%, 0 leaked
7 Labour market	Labour pre-read, projections, both statements	**–0.4 bp, neutral**	100%, 0 leaked
8 Financial conditions	Financial pre-read, base rates, both statements	**+3.4 bp hawkish**	100%, 0 leaked

One claim, as recorded

Communications analyst: the deterministic tone read scores the most recent policy communication as fully hawkish, with a unanimous hike vote and no dovish lexicon hits.

```
Quote: "score": 1.0   Item: signal.tone
Verified by code
```

The pre-read is our arithmetic over the two statements: 12 votes for, none against, eight hawkish words, no dovish ones. The analyst quoted it and code found the quote.

Live demo 4: what the analysts concluded

2 min

Open

- Analyst results view
- Analyst 1 (naive) card
- Analyst 4 (communications) card

Point at

- Analyst 1: cut 25 at 57%, numbers from memory flagged
- Analyst 4: the lean, grounded, leaked, watch-outs badges; one claim with its quote and the 'quote verified' mark
- Hover a badge for its definition

Say

- Every claim you see has a quote that code found character for character in the item named.
- The naive analyst still thinks the Fed is cutting. Same model, no evidence.

LIVE DEMO

The scorecard: eight of ten dimensions measured by code

S1 Repeatability measured by code	**S2** Groundedness measured by code	**S3** Leakage measured by code	**S4** Directional sanity measured by code	**S5** Noise invariance measured by code
S6 Paraphrase stability typed inputs only	**S7** Coherence measured by code	**S8** Reasoning quality not built: rubric grader	**S9** Auditability measured by code	**S10** Human-input discipline measured by code

Measured by code on the 28 October snapshot

- S4: 7 shocks, one input each, the pool must move the right way: 7 of 7
- S5: 5 irrelevant changes to the snapshot: 0.0 points moved
- S7: every ledger entry coherent and attributable
- S9: entries under the current spec recompute exactly, from the analyst runs they name

What it has already caught

- Four ledger entries stopped replaying after a model rule changed outside the hashed spec; amendment #5 put every rule inside the hash.
- A shock test was perturbing the wrong odds entry once two were on file; a red cell on the drift table pointed at it.
- Replay was reading the latest analyst runs instead of the runs the entry used; a rerun would have broken an honest entry. Fixed the same day, with a test.
- A verifier false positive (a zero written as 0 in the evidence); fixed, records re-scored without calling the model.

Live demo 5: the scorecard and the drift table

2 min

Open

- Scorecard view
- The snapshot dropdown: pick 24 October, then 26 October
- Expand 'what was checked' under S4
- Scroll to the over-time table

Point at

- The grid: red in the control columns, green in the judged columns
- S4's seven shocks with their deltas
- The over-time table: one row per snapshot, the marks never move

Say

- Analysts 1 and 2 are meant to fail. A red cell on the right would be a defect.
- This is the drift check. Two red cells have already pointed at defects and both are fixed.

LIVE DEMO

Live demo 6: the ledger and replay

1 min

Open

- Ledger view
- Press Verify ledger

Point at

- Every entry with its snapshot hash, spec hash and code version
- The replay result: chain intact, entries under the current spec recompute exactly
- Eight numbered amendments, each with its reason, each a new spec hash pinned in the tests

Say

- Nothing in this ledger can be edited or deleted without the chain breaking.
- Any entry can be recomputed from its snapshot. That is what auditable means here.

LIVE DEMO

What this does not claim, and what comes next

What this does not claim

One scored decision

28 October is the first. Everything before it is process evidence, which is the point, but calibration needs a record.

Weights are judgement

Market 0.6, Taylor 0.2, odds 0.1, base rates 0.1, the floor and the lean cap were set by reasoning and logged, never fitted.

The fitted model is outside

Probit v1 failed its gate and was reported, not tuned; v2 beats the base rate but carries a contamination caveat, so weight 0. About 130 usable decisions, 68% holds: no gradient-boosted anything.

The market is the anchor

At five weeks futures call most decisions. The claim is auditability and discipline, not a better hit rate.

What comes next, in order of value

1. ### The December window, same chain: running now

   The 9 December meeting is less certain (the market prices a hike at about three in four), so reading the evidence has more to add. The chain needed only a new target.

2. ### Pre-registered weight sets, scored live

   Three or four candidate sets written down now, all scored on every live decision; the live set changes only by amendment on that record.

3. ### An events agent for regime shifts

   1987, 1998, 2008, 2020: the Fed cut on financial stress, not the data the models read. A watch-out agent that scans for bank failures, funding stress and dislocations, and raises a watch-out every analyst must answer.

4. ### Reasoning quality and a second model family

   Twenty traces hand-graded, a model grader trusted only once it agrees; a second model family to show the harness, not the model, does the work.

Three things to take away

- The harness is the product. Same model, same evidence, same day: naive it wobbles and leaks; harnessed it is stable, quoted and checked.
- Judge the process. Eight decisions a year cannot prove skill; a pre-registered scorecard measured on every snapshot can show discipline and catch drift.
- The ask: scrutinise the scorecard, challenge a quote, and tell us which other decisions deserve the same chain. The 28 October call is on file; December is already running.